# Measures

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Remark

*Remark.* A measure assigns a size to each set in a collection of sets: how many elements it has, how long it is, or how likely it is. These notes build measures from $\sigma$-algebras and additivity, using the definitions on the Sets and Functions pages. One kind of measure, the probability measure (Definition 0.1), is the starting point of the Morrison Lab's probability notes (probability measure).

## $\sigma$-algebras

**Definition 0.1** ($\sigma$-algebra ($\sigma$-field, Borel field))**.** A **$\sigma$-algebra** [1] (also called a **$\sigma$-field** [2] or a **Borel field** [3], Definition 1.2.1) on a set $S$ is a collection $\mathcal{S}$ of subsets of $S$ that satisfies:

- $\mathcal{S}$ contains $S$ itself.
- For each set $A$ in $\mathcal{S}$, $\mathcal{S}$ contains its complement $S \setminus A$.
- For each sequence $A_1, A_2, \ldots$ of sets in $\mathcal{S}$, $\mathcal{S}$ contains their union $\bigcup_{i=1}^{\infty} A_i$.

Remark ("Borel" in "Borel field")

*Remark* 0.1 ("Borel" in "Borel field"). The "Borel" in "Borel field" is a different use of "Borel" from the Borel $\sigma$-algebra on $\mathbb{R}$, which is one particular $\sigma$-algebra.

**Example 0.1** ($\sigma$-algebras for die rolls)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), the collection of all subsets of $D$ is a $\sigma$-algebra on $D$, since complements and unions of subsets of $D$ are again subsets of $D$. So is the smaller collection

$$\{\emptyset, \{2,4,6\}, \{1,3,5\}, D\}$$

which contains $D$, the complement of each of its sets, and every union of its sets: for example, $\{2,4,6\} \cup \{1,3,5\} = D$.

**Example 0.2** (A collection that is not a $\sigma$-algebra)**.** The collection $\{\emptyset, \{1\}, D\}$ is not a $\sigma$-algebra on $D = \{1,2,3,4,5,6\}$: it contains $\{1\}$ but not its complement $D \setminus \{1\} = \{2,3,4,5,6\}$.

**Example 0.3** (A $\sigma$-algebra is closed under complements and countable unions)**.** A $\sigma$-algebra $\mathcal{S}$ on a set $S$ is closed under complements in $S$ and under unions of sequences of its sets, because Definition 0.1 requires both. The collection in Example 0.2 is not closed under complements.

**Theorem 0.1** (Closure properties of a $\sigma$-algebra)**.** *If $\mathcal{S}$ is a $\sigma$-algebra on a set $S$, then $\mathcal{S}$ contains:*

- *the empty set $\emptyset$;*
- *the union $A_1 \cup \cdots \cup A_n$ of any finitely many sets $A_1, \ldots, A_n$ in $\mathcal{S}$;*
- *the intersection $\bigcap_{i=1}^{\infty} A_i$ of any sequence $A_1, A_2, \ldots$ of sets in $\mathcal{S}$.*

*So $\mathcal{S}$ is closed under finite unions and under intersections of sequences of its sets.*

Proof

*Proof. Empty set.* $\mathcal{S}$ contains $S$, so it contains the complement of $S$, which is $S \setminus S = \emptyset$.
*Finite unions.* Extend $A_1, \ldots, A_n$ to a sequence by setting

$$\begin{aligned} A_{n+1} &= A_{n+2} \\ &= \cdots \\ &= \emptyset; \end{aligned}$$

each of these sets is in $\mathcal{S}$, by the first part. Then:

$$\begin{aligned} A_1 \cup \cdots \cup A_n &= A_1 \cup \cdots \cup A_n \cup \emptyset \cup \emptyset \cup \cdots && \text{(a union with } \emptyset \text{ adds no elements)} \\ &= \bigcup_{i=1}^{\infty} A_i && (A_i = \emptyset \text{ for } i > n) \end{aligned}$$

and $\mathcal{S}$ contains $\bigcup_{i=1}^{\infty} A_i$ by the union rule of Definition 0.1.
*Countable intersections.* Each $A_i$ is a subset of $S$, so an element of $S$ is outside $S \setminus A_i$ exactly when it is in $A_i$; that is, $S \setminus (S \setminus A_i) = A_i$. So, by De Morgan's laws:

$$S \setminus \bigcup_{i=1}^{\infty} (S \setminus A_i) = \bigcap_{i=1}^{\infty} (S \setminus (S \setminus A_i)) \quad \text{(De Morgan's laws)}$$
$$= \bigcap_{i=1}^{\infty} A_i \qquad (S \setminus (S \setminus A_i) = A_i)$$

Each $S \setminus A_i$ is in $\mathcal{S}$ by the complement rule, so their union is in $\mathcal{S}$ by the union rule, and the complement of that union is in $\mathcal{S}$ by the complement rule again. □

**Theorem 0.2** (All subsets form a $\sigma$-algebra)**.** *For any set $S$, the collection of all subsets of $S$ is a $\sigma$-algebra on $S$.*

Proof

*Proof.* Each condition of Definition 0.1 holds:

- $S$ is a subset of itself.
- For each subset $A$ of $S$, the complement $S \setminus A$ contains only elements of $S$, so it is a subset of $S$.
- For each sequence $A_1, A_2, \ldots$ of subsets of $S$, every element of $\bigcup_{i=1}^{\infty} A_i$ is in some $A_i$ and so in $S$; the union is a subset of $S$.

□

**Theorem 0.3** (An intersection of $\sigma$-algebras is a $\sigma$-algebra)**.** *Let $\mathcal{S}_j$ be a $\sigma$-algebra on a set $S$ for each $j$ in a nonempty set of indices $J$. Then their intersection $\bigcap_{j \in J} \mathcal{S}_j$, the collection of subsets of $S$ that are in every $\mathcal{S}_j$, is a $\sigma$-algebra on $S$.*

Proof

*Proof.* Each condition of Definition 0.1 holds:

- $S$ is in every $\mathcal{S}_j$, so it is in the intersection.
- If $A$ is in the intersection, then $A$ is in every $\mathcal{S}_j$, so its complement $S \setminus A$ is in every $\mathcal{S}_j$, and so it is in the intersection.
- If $A_1, A_2, \ldots$ are in the intersection, then they are in every $\mathcal{S}_j$, so their union $\bigcup_{i=1}^{\infty} A_i$ is in every $\mathcal{S}_j$, and so it is in the intersection.

□

**Definition 0.2** ($\sigma$-algebra generated by a collection)**.** Let $\mathcal{C}$ be a collection of subsets of a set $S$. The **$\sigma$-algebra generated by** $\mathcal{C}$, written $\sigma(\mathcal{C})$, is the intersection of all the $\sigma$-algebras on $S$ that contain every set in $\mathcal{C}$.

ℹ Remark (The smallest $\sigma$-algebra containing $\mathcal{C}$)

*Remark* 0.2 (The smallest $\sigma$-algebra containing $\mathcal{C}$). $\sigma(\mathcal{C})$ is a $\sigma$-algebra by Theorem 0.3 (the collection of all subsets of $S$ is one of the $\sigma$-algebras intersected, by Theorem 0.2, so there is at least one). It contains every set in $\mathcal{C}$, and it is contained in every other $\sigma$-algebra on $S$ that does, so it is the smallest $\sigma$-algebra on $S$ that contains $\mathcal{C}$.

**Example 0.4** (The $\sigma$-algebra generated by the even rolls)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$, let $\mathcal{C} = \{\{2, 4, 6\}\}$. Every $\sigma$-algebra on $D$ that contains $\{2, 4, 6\}$ must also contain $D$, the complement $\{1, 3, 5\}$, and $\emptyset$ (Definition 0.1 and Theorem 0.1). The collection $\{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\}$ is already a $\sigma$-algebra (Example 0.1), so it is $\sigma(\mathcal{C})$.

**Definition 0.3** (Borel $\sigma$-algebra)**.** The **Borel $\sigma$-algebra** on $\mathbb{R}$, written $\mathcal{B}$, is the $\sigma$-algebra generated by (Definition 0.2) the collection of all intervals.

**Example 0.5** (The Borel $\sigma$-algebra contains $\mathbb{R}$ and the half-lines)**.** The interval $(-\infty, 0)$ is in $\mathcal{B}$ by Definition 0.3. So is its complement $[0, \infty)$, because $\mathcal{B}$ is a $\sigma$-algebra (Definition 0.1). The set $\mathbb{R}$ is in $\mathcal{B}$ too: a $\sigma$-algebra on $\mathbb{R}$ contains $\mathbb{R}$ itself.

**Definition 0.4** (Borel set)**.** A **Borel set** [4] is a set in the Borel $\sigma$-algebra $\mathcal{B}$.

**Example 0.6** (Some Borel sets)**.**

- Every interval, such as $[0, 1]$, $(2, 5)$, or $[0, \infty)$, is a Borel set.
- A single point $\{a\} = [a, a]$ is an interval, so it is a Borel set.
- The integers are a Borel set: they are the union of the sequence of single points $\{0\}, \{1\}, \{-1\}, \{2\}, \{-2\}, \ldots$ (countable and uncountable sets lists them this way), and $\mathcal{B}$ contains that union by Definition 0.1.
- The complement $\mathbb{R} \setminus [0, 1]$ is a Borel set, because $\mathcal{B}$ contains the complement of each of its sets.

## Additivity

**Definition 0.1** (Finite additivity)**.** Let $\mathcal{S}$ be a $\sigma$-algebra on a set $S$. A function $\mu : \mathcal{S} \to [0, \infty]$, with values in the extended non-negative reals, is **finitely additive** [5] if, for every finite collection of pairwise disjoint sets $A_1, \ldots, A_n$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu(A_1 \cup \cdots \cup A_n) = \sum_{i=1}^{n} \mu(A_i)$$

**Example 0.1** (Counting elements is finitely additive)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), let $\mu(A) \stackrel{\text{def}}{=} |A|$, the number of elements of $A$ (cardinality), for each set $A$ in the $\sigma$-algebra of all subsets of $D$ (Example 0.1). The sets $\{1, 2\}$ and $\{5, 6\}$ are pairwise disjoint (low, middle and high die rolls), and:

$$\begin{aligned}
\mu(\{1,2\} \cup \{5,6\}) &= \mu(\{1,2,5,6\}) && \text{(take the union)} \\
&= 4 && \text{(count the elements)} \\
&= 2 + 2 && \text{(write 4 as a sum)} \\
&= \mu(\{1,2\}) + \mu(\{5,6\}) && \text{(count each set's elements)}
\end{aligned}$$

The same holds for any pairwise disjoint sets, because the sizes of disjoint sets add, so $\mu$ is finitely additive.

**Example 0.2** (Squaring the count is not finitely additive)**.** On the same $\sigma$-algebra, let $\nu(A) \stackrel{\text{def}}{=} |A|^2$. The sets $\{1\}$ and $\{2\}$ are pairwise disjoint, but

$$\begin{aligned}
\nu(\{1\} \cup \{2\}) &= \nu(\{1,2\}) && \text{(take the union)} \\
&= 2^2 \\
&= 4, && \text{(count, then square)} \\
\nu(\{1\}) + \nu(\{2\}) &= 1^2 + 1^2 \\
&= 2, && \text{(count each set, then square)}
\end{aligned}$$

and $4 \neq 2$, so $\nu$ is not finitely additive.

**Definition 0.2** (Countable additivity ($\sigma$-additivity))**.** Let $\mathcal{S}$ be a $\sigma$-algebra on a set $S$. A function $\mu : \mathcal{S} \to [0, \infty]$ is **countably additive** [5] (also called **$\sigma$-additive**) if, for every sequence of pairwise disjoint sets $A_1, A_2, \ldots$ in $\mathcal{S}$, the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

**Example 0.3** (Counting elements is countably additive)**.** For the counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.1, take any sequence of pairwise disjoint sets $A_1, A_2, \ldots$. No two of them share an element, and $D$ has only six elements, so at most six of the $A_i$ contain any elements; the rest are $\emptyset$, with $\mu(\emptyset) = 0$. The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.1) shows that those terms add up to $\mu$ of the union. So $\mu$ is countably additive.

Remark (The sum in countable additivity always has a value)

*Remark* 0.1 (The sum in countable additivity always has a value). By sums of non-negative terms, the right-hand side of the equation in Definition 0.2, $\sum_{i=1}^{\infty} \mu(A_i)$, always has a value, because each $\mu(A_i)$ is in $[0, \infty]$. For example, if $\mu(A_i) = 1/2^i$ for each $i$, the partial sums are $s_n = 1 - 1/2^n$, and the sum is 1; if $\mu(A_i) = 1$ for each $i$, the partial sums are $s_n = n$, and the sum is $\infty$.

**Lemma 0.1** (Countable additivity and the empty set)**.** *If $\mu$ is a countably additive function on a $\sigma$-algebra $\mathcal{S}$, then $\mu(\emptyset) = 0$ or $\mu(\emptyset) = \infty$.*

Proof

*Proof.* The set $\emptyset = S \setminus S$ is in $\mathcal{S}$, as the complement of $S$, and the sequence $\emptyset, \emptyset, \ldots$ is pairwise disjoint with union $\emptyset$, so:

$$\begin{aligned} \mu(\emptyset) &= \mu\left(\bigcup_{i=1}^{\infty} \emptyset\right) && \text{(the union of copies of } \emptyset \text{ is } \emptyset) \\ &= \sum_{i=1}^{\infty} \mu(\emptyset) && \text{(countable additivity)} \end{aligned}$$

If $\mu(\emptyset) = c$ for a finite $c > 0$, the right-hand side is $c + c + \cdots = \infty \neq c$, a contradiction. So $\mu(\emptyset)$ is 0 or $\infty$. □

**Example 0.4** (A countably additive function with $\mu(\emptyset) = 0$)**.** The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.3 is countably additive, and:

$$\begin{aligned} \mu(\emptyset) &= |\emptyset| && \text{(definition of } \mu\text{)} \\ &= 0 && (\emptyset \text{ has no elements}) \end{aligned}$$

**Example 0.5** (A countably additive function with $\mu(\emptyset) = \infty$)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$ (sets of die rolls), let $\mu(A) \stackrel{\text{def}}{=} \infty$ for every set $A$ in the $\sigma$-algebra of all subsets of $D$ (Example 0.1), including $A = \emptyset$. For any sequence of pairwise disjoint sets $A_1, A_2, \ldots$, the left-hand side of the countable additivity equation is:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \infty \quad \text{(definition of } \mu\text{)}$$

and the right-hand side, the limit of its partial sums (sums of non-negative terms), is:

$$\begin{aligned} \sum_{i=1}^{\infty} \mu(A_i) &= \infty + \infty + \cdots && \text{(definition of } \mu\text{)} \\ &= \infty && \text{(every partial sum is } \infty\text{)} \end{aligned}$$

The two sides agree, so $\mu$ is countably additive, and $\mu(\emptyset) = \infty$.

**i** Remark ($\mu(\emptyset) = 0$ is an extra requirement)

*Remark* 0.2 ($\mu(\emptyset) = 0$ is an extra requirement)*.* Example 0.4 and Example 0.5 show that both values allowed by Lemma 0.1 occur for countably additive functions. So $\mu(\emptyset) = 0$ is an extra requirement, not a consequence of countable additivity.

**Theorem 0.1** (Countable additivity implies finite additivity)**.** *If $\mu$ is a countably additive function on a $\sigma$-algebra $\mathcal{S}$, and $\mu(\emptyset) = 0$, then $\mu$ is finitely additive.*

**i** Proof

*Proof.* Let $A_1, \ldots, A_n$ be pairwise disjoint sets in $\mathcal{S}$, and extend them to a sequence by setting

$$\begin{aligned} A_{n+1} &= A_{n+2} \\ &= \cdots \\ &= \emptyset. \end{aligned}$$

The extended sequence is still pairwise disjoint, since $\emptyset$ shares no element with any set, and its union is $A_1 \cup \cdots \cup A_n$. So:

$$\begin{aligned}
\mu(A_1 \cup \cdots \cup A_n) &= \mu\left(\bigcup_{i=1}^{\infty} A_i\right) && (A_i = \emptyset \text{ for } i > n) \\
&= \sum_{i=1}^{\infty} \mu(A_i) && \text{(countable additivity)} \\
&= \sum_{i=1}^{n} \mu(A_i) + \sum_{i=n+1}^{\infty} \mu(\emptyset) && (A_i = \emptyset \text{ for } i > n) \\
&= \sum_{i=1}^{n} \mu(A_i) && (\mu(\emptyset) = 0)
\end{aligned}$$

□

**Example 0.6** (Finitely additive but not countably additive)**.** Let $S = \{0, 1, 2, ...\}$, with the $\sigma$-algebra of all subsets of $S$ (Theorem 0.2), and define:

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

*$\mu$ is finitely additive.* Let $A_1, \ldots, A_n$ be pairwise disjoint subsets of $S$. If every $A_i$ is finite, then so is their union, and:

$$\begin{aligned}
\mu(A_1 \cup \cdots \cup A_n) &= 0 && \text{(a finite union of finite sets is finite)} \\
&= \sum_{i=1}^{n} 0 && \text{(a sum of zeros is 0)} \\
&= \sum_{i=1}^{n} \mu(A_i) && \text{(each } A_i \text{ is finite)}
\end{aligned}$$

If some $A_k$ is infinite, then the union, which contains $A_k$, is infinite too, and:

$$\begin{aligned}
\mu(A_1 \cup \cdots \cup A_n) &= \infty && \text{(the union is infinite)} \\
&= \mu(A_k) + \sum_{i \in \{1,\ldots,n\} \setminus \{k\}} \mu(A_i) && (\mu(A_k) = \infty, \text{ and } \infty + x = \infty) \\
&= \sum_{i=1}^{n} \mu(A_i) && \text{(regroup the terms)}
\end{aligned}$$

*$\mu$ is not countably additive.* The single-element sets $\{0\}, \{1\}, \{2\}, \ldots$ are pairwise disjoint, and their union is $S$, which is infinite. So:

$$\begin{aligned}
\mu\left(\bigcup_{k=0}^{\infty} \{k\}\right) &= \mu(S) && \text{(the union is } S\text{)} \\
&= \infty && (S \text{ is infinite})
\end{aligned}$$

but:

$$\sum_{k=0}^{\infty} \mu(\{k\}) = 0 + 0 + \cdots \quad \text{(each } \{k\} \text{ is finite)}$$
$$= 0 \quad \text{(every partial sum is 0)}$$

Remark (Finite additivity does not imply countable additivity)

*Remark* 0.3 (Finite additivity does not imply countable additivity). Example 0.6 shows that the converse of Theorem 0.1 fails, even when $\mu(\emptyset) = 0$: a finitely additive function need not be countably additive [5]. The function $\mu$ in Example 0.6 has $\mu(\emptyset) = 0$, because $\emptyset$ is finite.

## Measures

**Definition 0.1** (Measure)**.** A **measure** on a set $S$ [6] with a $\sigma$-algebra $\mathcal{S}$ is a function $\mu : \mathcal{S} \to [0, \infty]$ that satisfies:

- $\mu(\emptyset) = 0$.
- $\mu$ is countably additive.

**Example 0.1** (Counting elements is a measure)**.** The counting function $\mu(A) \stackrel{\text{def}}{=} |A|$ of Example 0.1, defined on the $\sigma$-algebra of all subsets of $D$ (Example 0.1), takes values in $[0, \infty]$, is countably additive (Example 0.3), and gives $\mu(\emptyset) = 0$, so it is a measure.

**Corollary 0.1** (Measures are finitely additive)**.** *Every measure is finitely additive.*

Proof

*Proof.* A measure $\mu$ is countably additive and has $\mu(\emptyset) = 0$ (Definition 0.1), so Theorem 0.1 applies to it. □

**Definition 0.2** (Counting measure)**.** The **counting measure** [7] on a set $S$ is the measure on the $\sigma$-algebra of all subsets of $S$ that assigns each finite set its number of elements, and each infinite set the value $\infty$:

$$\mu(A) \stackrel{\text{def}}{=} \begin{cases} |A| & \text{if } A \text{ is finite} \\ \infty & \text{if } A \text{ is infinite} \end{cases}$$

**Example 0.2** (Counting measure on the non-negative integers)**.** For the counting measure $\mu$ on the non-negative integers $\{0, 1, 2, ...\}$, $\mu(\{0, 1, 2\}) = 3$, and the set of even numbers has $\mu(\{0, 2, 4, ...\}) = \infty$. The even numbers are the union of the pairwise disjoint sets $\{0\}, \{2\}, \{4\}, ...$, and countable additivity agrees:

$$\begin{aligned}\sum_{k=0}^{\infty} \mu(\{2k\}) &= 1 + 1 + \cdots \\ &= \infty.\end{aligned}$$

**Definition 0.3** (Measurable space)**.** A **measurable space** [8] $(S, \mathcal{S})$ is a set $S$ together with a $\sigma$-algebra $\mathcal{S}$ on $S$.

**Definition 0.4** (Measurable set)**.** In a measurable space $(S, \mathcal{S})$, the sets in $\mathcal{S}$ are its **measurable sets**.

**Example 0.3** (Measurable spaces of die rolls)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$:

- $(D, \mathcal{S})$, with $\mathcal{S}$ the collection of all subsets of $D$, is a measurable space (Example 0.1).
- $(D, \{\emptyset, \{2, 4, 6\}, \{1, 3, 5\}, D\})$ is also a measurable space. Its measurable sets are only those four; for example, $\{1\}$ is not one of them.

**Definition 0.5** (Measure space)**.** A **measure space** $(S, \mathcal{S}, \mu)$ is a measurable space together with a measure $\mu$ on $\mathcal{S}$.

**Example 0.4** (Measure space of die rolls)**.** For the die rolls $D = \{1, 2, 3, 4, 5, 6\}$, adding the counting function $\mu(A) = |A|$ of Example 0.1 to the measurable space $(D, \mathcal{S})$ of Example 0.3 makes $(D, \mathcal{S}, \mu)$ a measure space.

**Definition 0.6** (Lebesgue measure)**. Lebesgue measure** on $\mathbb{R}$ [9] is the measure $\lambda$ on the Borel $\sigma$-algebra $\mathcal{B}$ that assigns each interval its length: for real numbers $a \leq b$,

$$\begin{aligned}\lambda([a, b]) &= \lambda((a, b)) \\ &= \lambda([a, b)) \\ &= \lambda((a, b]) \\ &\stackrel{\text{def}}{=} b - a,\end{aligned}$$

and $\lambda(I) \stackrel{\text{def}}{=} \infty$ for each unbounded interval $I$.

Remark (Lebesgue measure exists)

*Remark* 0.1 (Lebesgue measure exists). Exactly one measure on $\mathcal{B}$ assigns every interval its length; constructing it takes more measure theory than these notes cover [9], which also extends $\lambda$ to a larger $\sigma$-algebra, the Lebesgue-measurable sets, with the same values on Borel sets.

**Definition 0.7** (Lebesgue measure on an interval)**. Lebesgue measure on an interval** $I$, such as $[0,1]$, is $\lambda$ (Definition 0.6) applied only to the Borel sets that are subsets of $I$.

**Example 0.5** (Lengths of some Borel sets)**.**

- $$\begin{aligned}\lambda([2,5]) &= 5-2\\ &= 3.\end{aligned}$$
- A single point has length zero:

$$\begin{aligned}\lambda(\{a\}) &= \lambda([a,a])\\ &= a-a\\ &= 0.\end{aligned}$$

- The integers are the union of the pairwise disjoint single points $\{0\},\{1\},\{-1\},\{2\},\{-2\},\ldots$ (Example 0.6), so countable additivity gives

$$\begin{aligned}\lambda(\mathbb{Z}) &= 0+0+\cdots\\ &= 0.\end{aligned}$$

- $\lambda([0,\infty)) = \infty$, because $[0,\infty)$ is unbounded.
- Lebesgue measure on $[0,1]$ gives the whole interval $[0,1]$ the value

$$\begin{aligned}\lambda([0,1]) &= 1-0\\ &= 1.\end{aligned}$$

Remark (Measures generalize size)

*Remark* 0.2 (Measures generalize size). A measure generalizes size: it can measure how many elements a set has, as in Definition 0.2, or how long a set of real numbers is, as Lebesgue measure (Definition 0.6) does, assigning each interval $[a,b]$ its length $b-a$.

For example, the counting measure on $\{0, 1, 2, ...\}$ gives $\{2, 3, 4, 5\}$ the value 4, while the Lebesgue measure of the interval $[2, 5]$ is $5 - 2 = 3$. Both can serve as the measures $\mu_1$ and $\mu_2$ in the Tonelli and Fubini theorems.

**Definition 0.8** (Measure zero, null set)**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. A set $A \in \mathcal{S}$ has **measure zero** [10] (also called a **null set** for $\mu$) if $\mu(A) = 0$.

**Example 0.6** (Sets of measure zero)**.**

- For a counting measure, only $\emptyset$ has measure zero: a nonempty set has at least one element, so its counting measure is at least 1.
- For Lebesgue measure, every single point $\{a\}$ and the integers $\mathbb{Z}$ have measure zero (Example 0.5), although neither set is empty.

**Definition 0.9** (Almost everywhere ($\mu$-almost everywhere, $\mu$-a.e.))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. A statement about the elements of $S$ holds **$\mu$-almost everywhere** (written **$\mu$-a.e.**, or **almost everywhere** when $\mu$ is clear from context) [11] if the set of elements for which it fails is a subset of a set of measure zero.

**Example 0.7** (Non-negative except on the integers)**.** Let $g : \mathbb{R} \to \mathbb{R}$ with $g(x) = -1$ when $x$ is an integer and $g(x) = x^2$ otherwise. The statement "$g(x) \geq 0$" fails exactly for $x \in \mathbb{Z}$, and $\lambda(\mathbb{Z}) = 0$ (Example 0.5), so $g \geq 0$ holds $\lambda$-almost everywhere, though not everywhere.
For a counting measure, only $\emptyset$ has measure zero (Example 0.6), so a statement holds almost everywhere only if it holds everywhere.

**Definition 0.10** (Finite measure)**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. The measure $\mu$ is **finite** if [12] $\mu(S) < \infty$.

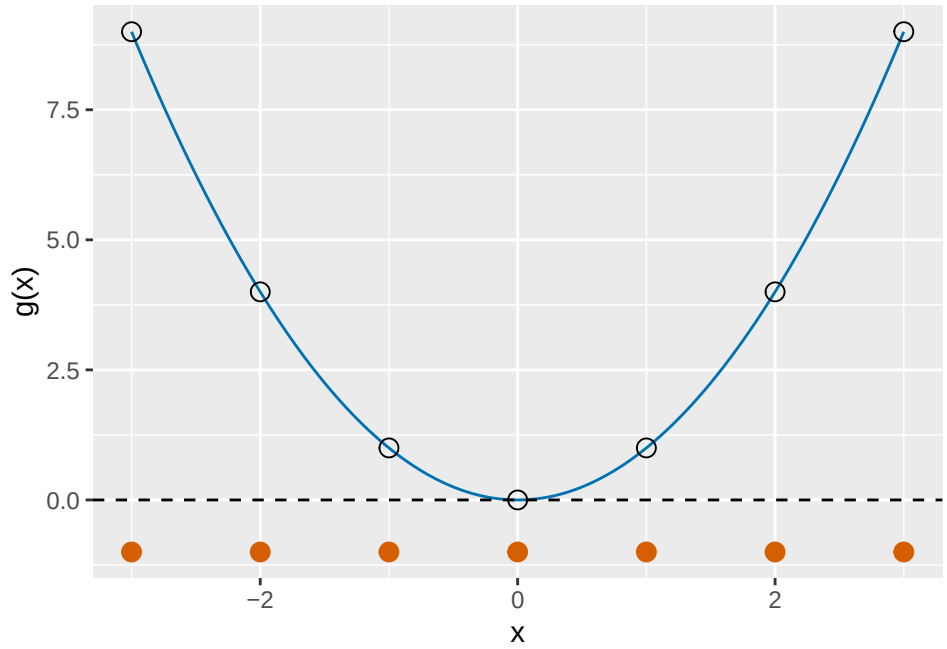
**Definition 0.11** ($\sigma$-finite measure)**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. The measure $\mu$ is **$\sigma$-finite** [13] if $S$ is the union $\bigcup_{i=1}^{\infty} A_i$ of a sequence of sets $A_1, A_2, ...$ in $\mathcal{S}$ with $\mu(A_i) < \infty$ for every $i$.

**Example 0.8** (Finite and $\sigma$-finite measures)**.**

- Every finite measure is $\sigma$-finite: take $A_1 = S$ and

**R**

```r
aeg_x_pf <- seq(-3, 3, length.out = 301)
aeg_curve_pf <- tibble::tibble(x = aeg_x_pf, y = aeg_x_pf^2)
aeg_ints_pf <- -3:3
aeg_open_pf <- tibble::tibble(x = aeg_ints_pf, y = aeg_ints_pf^2)
aeg_dots_pf <- tibble::tibble(x = aeg_ints_pf, y = -1)
aeg_aes_pf <- ggplot2::aes(x, y)
ggplot2::ggplot(aeg_curve_pf, aeg_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(data = aeg_open_pf, shape = 1, size = 3) +
  ggplot2::geom_point(data = aeg_dots_pf, colour = "#D55E00", size = 3) +
  ggplot2::geom_hline(yintercept = 0, linetype = "dashed") +
  ggplot2::labs(x = "x", y = "g(x)")
```

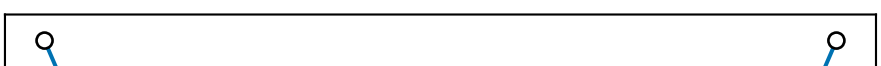



**Python**

```python
import matplotlib.pyplot as plt
import numpy as np

aeg_x_pf = np.linspace(-3, 3, 301)
aeg_ints_pf = np.arange(-3, 4)
fig, ax = plt.subplots(figsize=(5, 3.5), layout="constrained")
ax.plot(aeg_x_pf, aeg_x_pf**2, color="#0072B2")
ax.plot(aeg_ints_pf, aeg_ints_pf**2, linestyle="", marker="o",
        markerfacecolor="white", color="black")
ax.plot(aeg_ints_pf, -np.ones(7), linestyle="", marker="o", color="#D55E00")
ax.axhline(0, linestyle="dashed", color="black")
ax.set_xlabel("x")
ax.set_ylabel("g(x)")
plt.show()
```





$$\begin{aligned} A_2 &= A_3 \\ &= \cdots \\ &= \emptyset, \end{aligned}$$

with $\mu(A_1) = \mu(S) < \infty$ and $\mu(\emptyset) = 0$.

- The counting measure on the die rolls $D$ is finite, with $\mu(D) = 6$.
- The counting measure on $\{0, 1, 2, ...\}$ is not finite, but it is $\sigma$-finite: it is the union of the sets $A_i = \{0, 1, \dots, i\}$, each with $\mu(A_i) = i + 1 < \infty$.
- Lebesgue measure on $\mathbb{R}$ is not finite, since $\lambda(\mathbb{R}) = \infty$, but it is $\sigma$-finite: $\mathbb{R}$ is the union of the intervals $A_i = [-i, i]$, each with $\lambda(A_i) = 2i < \infty$.

## Probability measures

**Definition 0.1** (Probability measure)**.** A **probability measure** [14] on a measurable space $(\Omega, \mathcal{F})$ is a measure $\mathbb{P}$ on $\mathcal{F}$ that gives the whole set $\Omega$ the value 1:

$$\mathbb{P}(\Omega) = 1$$

**Example 0.1** (A fair die, and the unit interval)**.**

- For one roll of a fair die, the sample space is

$$\begin{aligned} \Omega &= D \\ &= \{1, 2, 3, 4, 5, 6\}, \end{aligned}$$

the events are all subsets of $D$, and $\mathbb{P}(A) \stackrel{\text{def}}{=} |A|/6$. $\mathbb{P}$ is the counting function $\mu(A) = |A|$ of Example 0.1 divided by 6. So $\mathbb{P}(\emptyset) = 0$, and $\mathbb{P}$ is countably additive, because each sum in Example 0.3 has at most six nonzero terms, and dividing a finite sum by 6 divides each term by 6. It is a probability measure, since

$$\begin{aligned} \mathbb{P}(D) &= 6/6 \\ &= 1. \end{aligned}$$

The outcome 3 is in the event "the roll is odd", $\{1, 3, 5\}$, which has

$$\begin{aligned}\mathbb{P}(\{1,3,5\}) &= 3/6 \\ &= 1/2.\end{aligned}$$

- Lebesgue measure on $[0,1]$ is a probability measure, because $\lambda([0,1]) = 1$ (Example 0.5).
- The counting measure on $D$ is not a probability measure, because it gives $D$ the value 6.

**Definition 0.2** (Probability space)**.** For a probability measure $\mathbb{P}$ (Definition 0.1) on a measurable space $(\Omega, \mathcal{F})$, $(\Omega, \mathcal{F}, \mathbb{P})$ is a **probability space**. [15]

**Definition 0.3** (Sample space)**.** In a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ (Definition 0.2), $\Omega$ is its **sample space**. [16]

**Definition 0.4** (Outcome)**.** In a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ (Definition 0.2), the elements of the sample space $\Omega$ (Definition 0.3) are **outcomes**. [17]

**Definition 0.5** (Event)**.** In a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ (Definition 0.2), the sets in $\mathcal{F}$ are **events**. [18]

Remark (Mutually exclusive events)

*Remark* 0.1 (Mutually exclusive events)*.* In probability, pairwise disjoint events are usually called **mutually exclusive**. For example, for one roll of a six-sided die, the events "the roll is even", $\{2,4,6\}$, and "the roll is odd", $\{1,3,5\}$, are mutually exclusive: no roll is both even and odd.

## Measurable functions and integrals

**Definition 0.1** (Measurable function)**.** Let $(S, \mathcal{S})$ be a measurable space. A function $f : S \to \mathbb{R}$ is **measurable** (with respect to $\mathcal{S}$) [19] if, for every real number $c$, the set of elements where $f$ is at most $c$ is measurable:

$$\{s \in S : f(s) \leq c\} \in \mathcal{S}$$

Remark (Measurable functions in general)

*Remark* 0.1 (Measurable functions in general). Other sources define a function $f : S \to T$ between measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ to be measurable when $\{s \in S : f(s) \in B\} \in \mathcal{S}$ for every $B \in \mathcal{T}$. For $T = \mathbb{R}$ with the Borel $\sigma$-algebra, that definition is equivalent to Definition 0.1. For example, with $B = (-\infty, c]$, the general definition requires $\{s \in S : f(s) \le c\} \in \mathcal{S}$, which is the condition in Definition 0.1.

**Example 0.1** (Measurable and not measurable)**.** On the die rolls $D$, take the $\sigma$-algebra $\mathcal{S} = \{\emptyset, \{2,4,6\}, \{1,3,5\}, D\}$ of Example 0.1.

- Let $f(x) = 1$ for even $x$ and $f(x) = 0$ for odd $x$. The set $\{x \in D : f(x) \le c\}$ is $\emptyset$ when $c < 0$, $\{1,3,5\}$ when $0 \le c < 1$, and $D$ when $c \ge 1$. All three are in $\mathcal{S}$, so $f$ is measurable.
- Let $g(x) = x$. Then $\{x \in D : g(x) \le 1\} = \{1\}$, which is not in $\mathcal{S}$, so $g$ is not measurable.

With the $\sigma$-algebra of all subsets of $D$ instead, every set $\{x \in D : g(x) \le c\}$ is a subset of $D$, so every function $D \to \mathbb{R}$, including $g$, is measurable.

**Definition 0.2** (Integral of a non-negative function)**.** Let $(S, \mathcal{S}, \mu)$ be a measure space, and let $f : S \to [0, \infty)$ be measurable. The **integral** of $f$ with respect to $\mu$ is [20]

$$\int_S f \, d\mu \stackrel{\text{def}}{=} \sup \left\{ \sum_{i=1}^{n} \Big( \inf_{s \in A_i} f(s) \Big) \mu(A_i) \right\}$$

where the supremum is over all ways of writing $S$ as the union of finitely many nonempty pairwise disjoint sets $A_1, \ldots, A_n$ in $\mathcal{S}$, and each $\inf_{s \in A_i} f(s)$ is the infimum of the values of $f$ on $A_i$. In the sums, $0 \cdot \infty \stackrel{\text{def}}{=} 0$, and $x \cdot \infty \stackrel{\text{def}}{=} \infty$ for $x > 0$; the supremum is $\infty$ if some sum is $\infty$. The integral is also written $\int_S f(s) \, d\mu(s)$.
[2, Sec. 15]

Remark (Each sum is a lower estimate)

*Remark* 0.2 (Each sum is a lower estimate). Each sum in Definition 0.2 splits $S$ into pieces and weights each piece $A_i$ by the smallest value $f$ approaches on it, so each sum is at most the integral. For Lebesgue measure on a closed interval $[a, b]$ and a function $f \ge 0$ that has a Riemann integral on $[a, b]$, the two integrals agree: $\int_{[a,b]} f \, d\lambda = \int_a^b f(x) \, dx$ [20].

**Example 0.2** (Integrals with respect to counting measure)**.** Let $\mu$ be the counting measure on the die rolls $D$, and let $g : D \to [0, \infty)$. Every function on $D$ is measurable (Example 0.1).

*The integral is the sum of the values.* For any split of $D$ into nonempty pairwise disjoint sets $A_1, \dots, A_n$:

$$\begin{aligned}
\sum_{i=1}^{n} \Big( \inf_{x \in A_i} g(x) \Big) \mu(A_i) &= \sum_{i=1}^{n} \sum_{x \in A_i} \inf_{y \in A_i} g(y) && (\mu(A_i) \text{ counts the elements of } A_i) \\
&\le \sum_{i=1}^{n} \sum_{x \in A_i} g(x) && (\text{an infimum is at most each value}) \\
&= \sum_{x \in D} g(x) && (\text{each } x \in D \text{ is in exactly one } A_i)
\end{aligned}$$

The split into the six single points $\{1\}, \dots, \{6\}$ gives exactly $\sum_{x \in D} g(x)$, so the supremum is $\int_D g \, d\mu = \sum_{x \in D} g(x)$. For example, with $g(x) = x$,

$$\begin{aligned}
\int_D g \, d\mu &= 1 + 2 + \cdots + 6 \\
&= 21.
\end{aligned}$$

*An infinite integral.* For the counting measure $\mu$ on $\mathbb{N}$ and the constant function $h(n) = 1$, the split with one set, $A_1 = \mathbb{N}$, gives the sum

$$\begin{aligned}
1 \cdot \mu(\mathbb{N}) &= 1 \cdot \infty \\
&= \infty,
\end{aligned}$$

so $\int_{\mathbb{N}} h \, d\mu = \infty$.

ℹ Remark (A function in terms of its positive and negative parts)

*Remark* 0.3 (A function in terms of its positive and negative parts). For a function $f : S \to \mathbb{R}$, with the positive part and negative part, $f = f^+ - f^-$ and $|f| = f^+ + f^-$. ($f^+$, $f^-$, and $|f|$ are measurable when $f$ is [2], Theorem 13.3.)

**Definition 0.3** (Integral of a real-valued function)**.** Let $(S, \mathcal{S}, \mu)$ be a measure space, and let $f : S \to \mathbb{R}$ be measurable. Let $f^+$ and $f^-$ be the positive part and negative part of $f$. The **integral** of $f$ with respect to $\mu$ is

$$\int_S f \, d\mu \stackrel{\text{def}}{=} \int_S f^+ \, d\mu - \int_S f^- \, d\mu,$$

using Definition 0.2 for each term, when at least one of the two terms is finite; otherwise,

the integral of $f$ is undefined.
[2, Sec. 15]

**Example 0.3** (An integral with positive and negative parts)**.** For the counting measure $\mu$ on the die rolls $D$, let $f(x) = x - 3$. Its values at $1, \dots, 6$ are $-2, -1, 0, 1, 2, 3$, so:

- $f^+$ has values $0, 0, 0, 1, 2, 3$, and $\int_D f^+ \, d\mu = 6$ by Example 0.2;
- $f^-$ has values $2, 1, 0, 0, 0, 0$, and $\int_D f^- \, d\mu = 3$;
- $$\begin{aligned} \int_D f \, d\mu &= 6 - 3 \\ &= 3, \end{aligned}$$

  which is also

  $$\begin{aligned} \sum_{x \in D} (x - 3) &= 21 - 18 \\ &= 3. \end{aligned}$$

**Definition 0.4** (Absolutely integrable function, integrable function)**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. A measurable function $f : S \to \mathbb{R}$ is **absolutely integrable** with respect to $\mu$ (often just called **integrable**) if the integral of its absolute value is finite:

$$\int_S |f| \, d\mu < \infty$$

The property itself is called absolute integrability.

Remark (An absolutely integrable function has a finite integral)

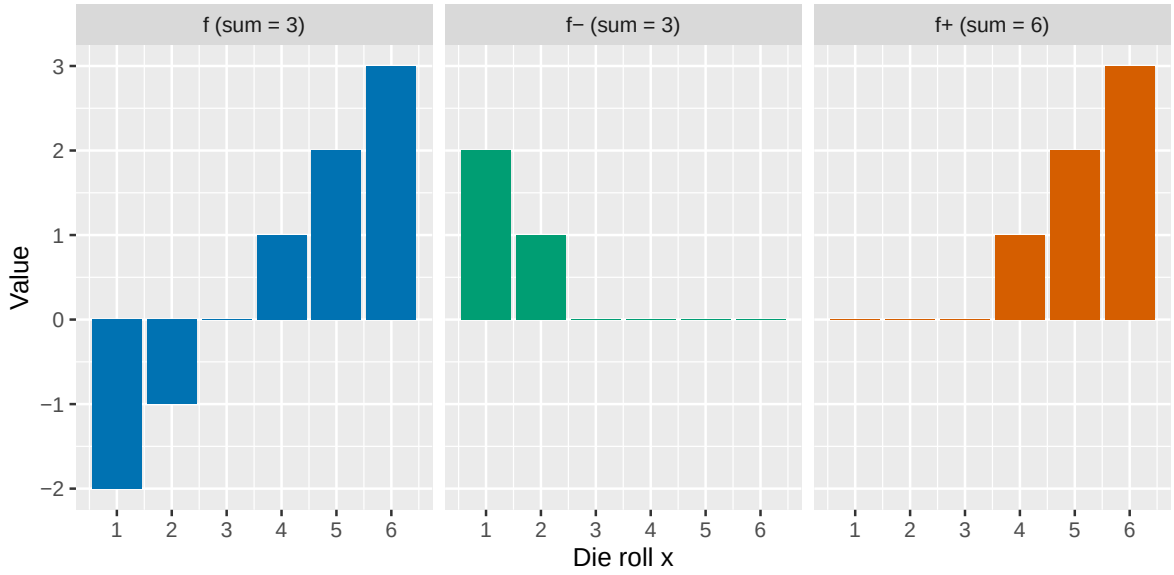
*Remark* 0.4 (An absolutely integrable function has a finite integral). If $f$ is absolutely integrable, then $\int_S f^+ \, d\mu$ and $\int_S f^- \, d\mu$ are both finite, because $f^+ \le |f|$ and $f^- \le |f|$, so each sum in Definition 0.2 for $f^+$ or $f^-$ is at most the matching sum for $|f|$. So $\int_S f \, d\mu$ (Definition 0.3) is a real number.

**Example 0.4** (Absolutely integrable and not)**.**

- For the counting measure on the die rolls $D$, $f(x) = x - 3$ (Example 0.3) is absolutely integrable:

**R**

```
pnp_rolls_pf <- 1:6
pnp_f_pf <- pnp_rolls_pf - 3
pnp_plus_pf <- pmax(pnp_f_pf, 0)
pnp_minus_pf <- pmax(-pnp_f_pf, 0)
pnp_f_tbl_pf <- tibble::tibble(
  x = pnp_rolls_pf, value = pnp_f_pf, panel = "f (sum = 3)"
)
pnp_plus_tbl_pf <- tibble::tibble(
  x = pnp_rolls_pf, value = pnp_plus_pf, panel = "f+ (sum = 6)"
)
pnp_minus_tbl_pf <- tibble::tibble(
  x = pnp_rolls_pf, value = pnp_minus_pf, panel = "f- (sum = 3)"
)
pnp_bars_pf <- dplyr::bind_rows(pnp_f_tbl_pf, pnp_plus_tbl_pf, pnp_minus_tbl_pf)
pnp_aes_pf <- ggplot2::aes(x, value, fill = panel)
pnp_colours_pf <- c("#0072B2", "#009E73", "#D55E00")
ggplot2::ggplot(pnp_bars_pf, pnp_aes_pf) +
  ggplot2::geom_col() +
  ggplot2::facet_wrap(ggplot2::vars(panel)) +
  ggplot2::scale_fill_manual(values = pnp_colours_pf) +
  ggplot2::scale_x_continuous(breaks = pnp_rolls_pf) +
  ggplot2::guides(fill = "none") +
  ggplot2::labs(x = "Die roll x", y = "Value")
```




**Python**

```
import matplotlib.pyplot as plt
import numpy as np

pnp_rolls_pf = np.arange(1, 7)
pnp_f_pf = pnp_rolls_pf - 3
pnp_panels_pf = [
    ("f (sum = 3)", pnp_f_pf, "#0072B2"),
    ("f+ (sum = 6)", np.maximum(pnp_f_pf, 0), "#009E73"),
```

$$\begin{aligned}\int_D |f|\,d\mu &= 2+1+0+1+2+3\\ &= 9\\ &< \infty\end{aligned}$$

by Example 0.2.

- For the counting measure on $\mathbb{N}$, the constant function $h(n) = 1$ is not absolutely integrable, because

$$\begin{aligned}\int_{\mathbb{N}} |h|\,d\mu &= \int_{\mathbb{N}} h\,d\mu\\ &= \infty\end{aligned}$$

(Example 0.2).

## $L^p$ spaces

**Definition 0.1** ($L^p$ norm ($p$-norm of a function))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. For any real number $p \geq 1$, the $L^p$ **norm** of a measurable function $f : S \to \mathbb{R}$ is:

$$\|f\|_p \stackrel{\text{def}}{=} \left(\int_S |f|^p\,d\mu\right)^{1/p}.$$

**Example 0.1** (The $L^1$ and $L^2$ norms of $x \mapsto x$ on the unit interval)**.** Let $S = [0, 1]$ with the Lebesgue measure $\mu$, and let $f(x) = x$. By Definition 0.1,

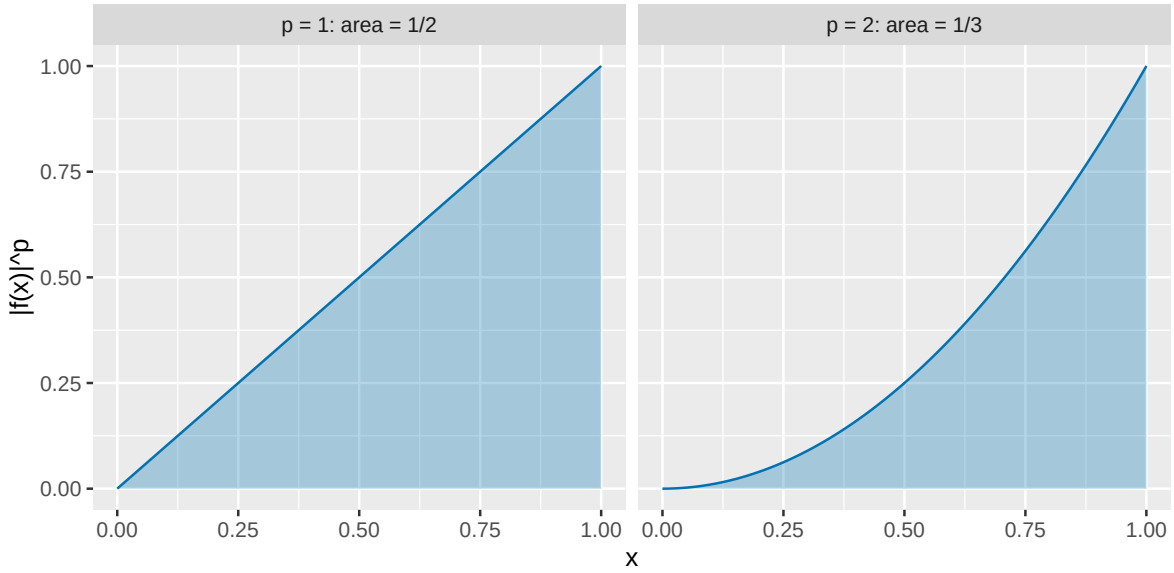
$$\begin{aligned}\|f\|_1 &= \int_0^1 x\,dx\\ &= \frac{1}{2},\end{aligned}$$

and

$$\begin{aligned}\|f\|_2 &= \left(\int_0^1 x^2\,dx\right)^{1/2}\\ &= \left(\frac{1}{3}\right)^{1/2}.\end{aligned}$$

**R**

```r
lpna_x_pf <- seq(0, 1, length.out = 201)
lpna_p1_pf <- tibble::tibble(
  x = lpna_x_pf, y = lpna_x_pf, panel = "p = 1: area = 1/2"
)
lpna_sq_pf <- lpna_x_pf^2
lpna_p2_pf <- tibble::tibble(
  x = lpna_x_pf, y = lpna_sq_pf, panel = "p = 2: area = 1/3"
)
lpna_curves_pf <- dplyr::bind_rows(lpna_p1_pf, lpna_p2_pf)
lpna_aes_pf <- ggplot2::aes(x, y)
ggplot2::ggplot(lpna_curves_pf, lpna_aes_pf) +
  ggplot2::geom_area(fill = "#0072B2", alpha = 0.3) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::facet_wrap(ggplot2::vars(panel)) +
  ggplot2::labs(x = "x", y = "|f(x)|^p")
```




**Python**

```python
import matplotlib.pyplot as plt
import numpy as np

lpna_x_pf = np.linspace(0, 1, 201)
fig, axes = plt.subplots(1, 2, figsize=(7, 3.5), sharey=True,
                         layout="constrained")
for ax, p, title in zip(axes, [1, 2], ["p = 1: area = 1/2", "p = 2: area = 1/3"]):
    ax.fill_between(lpna_x_pf, lpna_x_pf**p, color="#0072B2", alpha=0.3)
    ax.plot(lpna_x_pf, lpna_x_pf**p, color="#0072B2")
    ax.set_title(title)
    ax.set_xlabel("x")
axes[0].set_ylabel("|f(x)|^p")
plt.show()
```





**Definition 0.2** ($\mathcal{L}^p$ space ($\mathcal{L}^p(S, \mathcal{S}, \mu)$))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. Let $p \geq 1$ be a real number. The $\mathcal{L}^p$ **space** $\mathcal{L}^p(S, \mathcal{S}, \mu)$ is the set of all measurable functions $f : S \to \mathbb{R}$ whose $L^p$ norm is less than $\infty$: $\|f\|_p < \infty$.

**Example 0.2** (An $L^1$ function not in $L^2$)**.** Consider the interval $(0, 1)$ with the Borel $\sigma$-algebra (Definition 0.3) and Lebesgue measure $\mu$ (Definition 0.6). Define $f : (0, 1) \to \mathbb{R}$ by $f(x) = x^{-1/2}$.

1. For $p = 1$:
$$\begin{aligned} \|f\|_1 &= \int_0^1 x^{-1/2}\, dx \\ &= \left[2x^{1/2}\right]_0^1 \\ &= 2 \\ &< \infty, \end{aligned}$$
so $f \in \mathcal{L}^1((0, 1), \mu)$.
2. For $p = 2$:
$$\begin{aligned} |f(x)|^2 &= \left(x^{-1/2}\right)^2 \\ &= \frac{1}{x}. \end{aligned}$$
The integral is:
$$\begin{aligned} \int_0^1 \frac{1}{x}\, dx &= \lim_{\varepsilon \to 0^+} [\ln x]_\varepsilon^1 \\ &= \lim_{\varepsilon \to 0^+} (-\ln \varepsilon) \\ &= \infty. \end{aligned}$$
So $\|f\|_2 = \infty$, so $f \notin \mathcal{L}^2((0, 1), \mu)$.

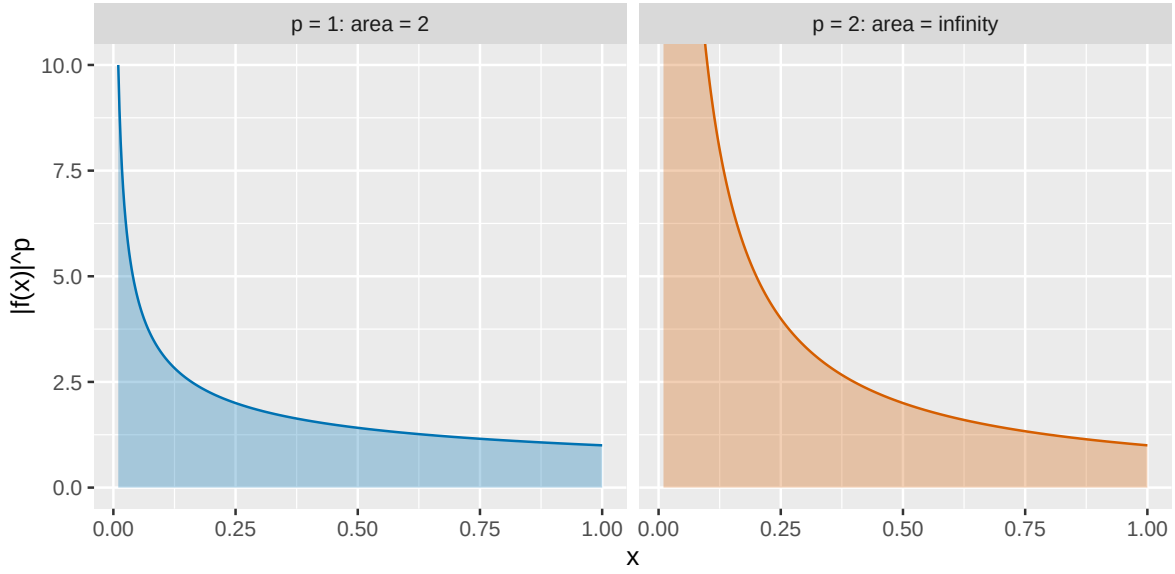
This shows that $\mathcal{L}^2 \subset \mathcal{L}^1$ on sets of finite measure: a larger $p$ imposes stricter requirements near singularities.

**Definition 0.3** (Equal almost everywhere ($f = g$ $\mu$-a.e.))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. Let $f, g : S \to \mathbb{R}$ be measurable. The functions $f$ and $g$ are **equal almost everywhere**, written $f = g$ $\mu$-a.e., if $f(s) = g(s)$ holds almost everywhere.

**Example 0.3** (A function that differs from 0 at one point)**.** Let $S = \mathbb{R}$ with the Lebesgue measure $\mu$. Let $f(s) = 1$ if $s = 0$, and $f(s) = 0$ otherwise. The set $\{s : f(s) \neq 0\} = \{0\}$ has measure zero, so $f = 0$ almost everywhere by Definition 0.3.

**R**

```r
lpss_x_pf <- seq(0.01, 1, length.out = 400)
lpss_root_pf <- lpss_x_pf^(-1 / 2)
lpss_p1_pf <- tibble::tibble(
  x = lpss_x_pf, y = lpss_root_pf, panel = "p = 1: area = 2"
)
lpss_inv_pf <- 1 / lpss_x_pf
lpss_p2_pf <- tibble::tibble(
  x = lpss_x_pf, y = lpss_inv_pf, panel = "p = 2: area = infinity"
)
lpss_curves_pf <- dplyr::bind_rows(lpss_p1_pf, lpss_p2_pf)
lpss_aes_pf <- ggplot2::aes(x, y, fill = panel, colour = panel)
lpss_colours_pf <- c("#0072B2", "#D55E00")
lpss_ylim_pf <- c(0, 10)
ggplot2::ggplot(lpss_curves_pf, lpss_aes_pf) +
  ggplot2::geom_area(alpha = 0.3, colour = NA) +
  ggplot2::geom_line() +
  ggplot2::facet_wrap(ggplot2::vars(panel)) +
  ggplot2::scale_fill_manual(values = lpss_colours_pf) +
  ggplot2::scale_colour_manual(values = lpss_colours_pf) +
  ggplot2::coord_cartesian(ylim = lpss_ylim_pf) +
  ggplot2::guides(fill = "none", colour = "none") +
  ggplot2::labs(x = "x", y = "|f(x)|^p")
```




**Python**

```python
import matplotlib.pyplot as plt
import numpy as np

lpss_x_pf = np.linspace(0.01, 1, 400)
lpss_curves_pf = [
    ("p = 1: area = 2", lpss_x_pf ** (-1 / 2), "#0072B2"),
    ("p = 2: area = infinity", 1 / lpss_x_pf, "#D55E00"),
]
fig, axes = plt.subplots(1, 2, figsize=(7, 3.5), sharey=True,
```

**Definition 0.4** ($L^p$ space ($L^p(S, \mathcal{S}, \mu)$, $L^p(\mu)$))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. Let $p \geq 1$ be a real number. For each $f$ in the $\mathcal{L}^p$ space, let $[f]$ be the set of all functions in $\mathcal{L}^p(S, \mathcal{S}, \mu)$ that are equal almost everywhere to $f$. The $L^p$ **space** $L^p(S, \mathcal{S}, \mu)$ (or simply $L^p(\mu)$) is the set of all such sets $[f]$, with the norm $\|[f]\|_p \stackrel{\text{def}}{=} \|f\|_p$ ($L^p$ norm).

**Example 0.4** (The zero element of $L^p$)**.** Let $f$ be the function in Example 0.3, which is 1 at $s = 0$ and 0 elsewhere. Then $f = 0$ almost everywhere, so $[f] = [0]$. Also $\|f\|_p = 0$ although $f \neq 0$, so $\|f\|_p = 0$ can hold for a nonzero element of $\mathcal{L}^p$; on $L^p(\mu)$, $\|[f]\|_p = 0$ only for $[f] = [0]$, as a norm requires.

**Definition 0.5** (Essential supremum norm ($L^\infty$ norm))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. The **essential supremum norm** (or $L^\infty$ **norm**) of a measurable function $f : S \to \mathbb{R}$ is:

$$\|f\|_\infty \stackrel{\text{def}}{=} \inf\{C \geq 0 : \mu(\{s \in S : |f(s)| > C\}) = 0\}.$$

**Example 0.5** (The essential supremum norm ignores a null set)**.** Let $S = [0, 1]$ with the Lebesgue measure $\mu$. Let $f(s) = 5$ if $s = 0$ and $f(s) = s$ otherwise. Then $\sup_s |f(s)| = 5$. For $C \geq 1$, the set $\{s : |f(s)| > C\}$ is empty or $\{0\}$, so it has measure zero, and for $C < 1$ it has positive measure. By Definition 0.5, $\|f\|_\infty = 1$.

**Definition 0.6** ($\mathcal{L}^\infty$ space ($\mathcal{L}^\infty(S, \mathcal{S}, \mu)$))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. The $\mathcal{L}^\infty$ **space** $\mathcal{L}^\infty(S, \mathcal{S}, \mu)$ is the set of all measurable functions $f : S \to \mathbb{R}$ whose essential supremum norm is less than $\infty$: $\|f\|_\infty < \infty$.

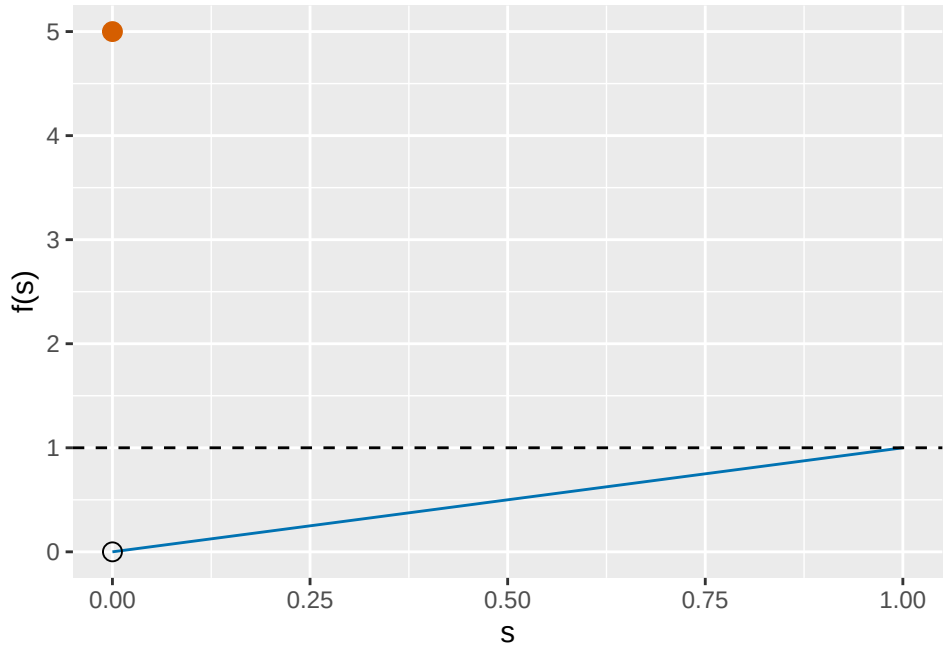
**Example 0.6** (Bounded and unbounded functions)**.** Let $S = (0, 1)$ with the Lebesgue measure $\mu$. The function $f(s) = s$ has $\|f\|_\infty = 1 < \infty$, so $f \in \mathcal{L}^\infty(S, \mathcal{S}, \mu)$ by Definition 0.6. The function $g(s) = 1/s$ exceeds every $C$ on the set $(0, 1/C)$ of positive measure, so $\|g\|_\infty = \infty$ and $g \notin \mathcal{L}^\infty(S, \mathcal{S}, \mu)$.

**Definition 0.7** ($L^\infty$ space ($L^\infty(S, \mathcal{S}, \mu)$, $L^\infty(\mu)$))**.** Let $(S, \mathcal{S}, \mu)$ be a measure space. For each $f$ in the $\mathcal{L}^\infty$ space, let $[f]$ be the set of all functions in $\mathcal{L}^\infty(S, \mathcal{S}, \mu)$ that are equal almost everywhere to $f$. The $L^\infty$ **space** $L^\infty(S, \mathcal{S}, \mu)$ (or simply $L^\infty(\mu)$) is the set of all such sets $[f]$, with the norm $\|[f]\|_\infty \stackrel{\text{def}}{=} \|f\|_\infty$ (essential supremum norm).

**Example 0.7** (Bounded functions equal almost everywhere)**.** Let $f$ be the function in Example 0.5 and $h(s) = s$ on $[0, 1]$. Then $f = h$ almost everywhere (Definition 0.3), so

**R**

```r
esn_x_pf <- seq(0, 1, length.out = 101)
esn_line_pf <- tibble::tibble(x = esn_x_pf, y = esn_x_pf)
esn_open_pf <- tibble::tibble(x = 0, y = 0)
esn_spike_pf <- tibble::tibble(x = 0, y = 5)
esn_aes_pf <- ggplot2::aes(x, y)
ggplot2::ggplot(esn_line_pf, esn_aes_pf) +
  ggplot2::geom_line(colour = "#0072B2") +
  ggplot2::geom_point(data = esn_open_pf, shape = 1, size = 3) +
  ggplot2::geom_point(data = esn_spike_pf, colour = "#D55E00", size = 3) +
  ggplot2::geom_hline(yintercept = 1, linetype = "dashed") +
  ggplot2::labs(x = "s", y = "f(s)")
```




**Python**

```python
import matplotlib.pyplot as plt
import numpy as np

esn_x_pf = np.linspace(0, 1, 101)
fig, ax = plt.subplots(figsize=(5, 3.5), layout="constrained")
ax.plot(esn_x_pf, esn_x_pf, color="#0072B2")
ax.plot(0, 0, marker="o", markerfacecolor="white", color="black")
ax.plot(0, 5, marker="o", color="#D55E00")
ax.axhline(1, linestyle="dashed", color="black")
ax.set_xlabel("s")
ax.set_ylabel("f(s)")
plt.show()
```





$[f] = [h]$ in $L^\infty(\mu)$, and $\|[f]\|_\infty = \|h\|_\infty = 1$.

**Theorem 0.1** (Holder's inequality)**.** *Let $(S, \mathcal{S}, \mu)$ be a measure space, and let $p, q \in [1, \infty]$ be conjugate exponents satisfying $\frac{1}{p} + \frac{1}{q} = 1$ (with the convention that $q = \infty$ when $p = 1$, and $q = 1$ when $p = \infty$). If $f \in L^p(\mu)$ and $g \in L^q(\mu)$ (Definition 0.4), then their product $fg$ is absolutely integrable ($fg \in L^1(\mu)$) and satisfies:*

$$\|fg\|_1 = \int_S |fg| \, d\mu \leq \|f\|_p \, \|g\|_q.$$

*When*

$$\begin{aligned} p &= q \\ &= 2, \end{aligned}$$

*this inequality is the Cauchy-Schwarz inequality for integrals.*

**Theorem 0.2** (Minkowski's inequality ($L^p$ triangle inequality))**.** *Let $(S, \mathcal{S}, \mu)$ be a measure space and let $1 \leq p \leq \infty$. For any functions $f, g \in L^p(\mu)$ (Definition 0.4), their sum $f + g$ is in $L^p(\mu)$ and satisfies:*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

*Together with absolute homogeneity $\|cf\|_p = |c| \, \|f\|_p$ and definiteness, Minkowski's inequality establishes that $L^p(\mu)$ satisfies all axioms of a normed vector space.*

**Example 0.8** (Applying Holder's inequality to finite sums and expectations)**.**

1. **Finite vectors (Cauchy-Schwarz and taxicab bounds)**: Equipping a finite set with the counting measure, Holder's inequality for conjugate exponents $p, q$ gives:

$$\sum_{i=1}^n |x_i y_i| \leq \left( \sum_{i=1}^n |x_i|^p \right)^{1/p} \left( \sum_{i=1}^n |y_i|^q \right)^{1/q}.$$

   Setting $p = 1$, $q = \infty$ gives $\sum_{i=1}^n |x_i y_i| \leq \|\underset{\sim}{x}\|_1 \, \|\underset{\sim}{y}\|_\infty$. Setting

$$\begin{aligned} p &= q \\ &= 2 \end{aligned}$$

   gives the standard vector Cauchy-Schwarz inequality.

2. **Covariance bound for random variables**: On a probability space with measure $\mathbb{P}$ (Definition 0.1), taking

$$
\begin{aligned}
p &= q \\
&= 2
\end{aligned}
$$

gives $\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2]}\sqrt{\mathbb{E}[Y^2]}$, guaranteeing that the covariance between any two square-integrable random variables is finite.

Remark ($L^p$ spaces and norms in probability and machine learning)

*Remark* 0.1 ($L^p$ spaces and norms in probability and machine learning). In probability, statistics, and machine learning, $L^p$ spaces structure the geometry of random variables, estimators, and loss functions:

1. **Random variables, moments, and inclusion of $L^p$ spaces**: When $\mu = \mathbb{P}$ is a probability measure (Definition 0.1) with total measure $\mathbb{P}(S) = 1$, an $L^p$ function is a random variable $X$ with finite $p$-th moment:

   $$\|X\|_p = (\mathbb{E}[|X|^p])^{1/p}.$$

   By Jensen's inequality, if $1 \leq p \leq q \leq \infty$, then $\|X\|_p \leq \|X\|_q$, so $L^q(P) \subseteq L^p(P)$. For instance, every random variable with finite variance ($X \in L^2$) has finite expectation ($X \in L^1$).

2. **The unique Hilbert space $L^2$ and conditional expectation**: Among all $L^p$ spaces, only $L^2(\mu)$ is an inner product space, equipped with the inner product $\langle f, g\rangle \overset{\text{def}}{=} \int_S fg\, d\mu$ (all other $L^p$ spaces fail the parallelogram law). The completeness of $L^2$ makes it a Hilbert space. In regression, the conditional expectation $\mathbb{E}[Y \mid X]$ is the orthogonal projection of $Y \in L^2$ onto the closed subspace of $L^2$ functions measurable with respect to $X$, minimizing the population mean squared error (pMSE) $\|Y - g(X)\|_2^2$.

3. **Loss functions, estimation, and robustness**: The choice of $p$ in $L^p$ norms defines loss functions for optimization and regression:

   - **$L^2$ loss (population mean squared error (pMSE), or sample mean squared error (sMSE) on a sample)**: penalizes large errors quadratically, yielding the conditional mean; sensitive to outliers.
   - **$L^1$ loss (population mean absolute error (pMAE), or sample mean absolute error (sMAE) on a sample)**: penalizes errors linearly, yielding the conditional median; robust to heavy tails and outliers.
   - **$L^\infty$ loss (Chebyshev criterion)**: minimizes the worst-case maximum error across all instances.

## Product measures

**Definition 0.1** (Measurable rectangle)**.** Let $(S, \mathcal{S})$ and $(T, \mathcal{T})$ be measurable spaces. A **measurable rectangle** is a Cartesian product $A \times B$ with $A \in \mathcal{S}$ and $B \in \mathcal{T}$.

**Definition 0.2** (Product $\sigma$-algebra)**.** For the measurable spaces $(S, \mathcal{S})$ and $(T, \mathcal{T})$ of Definition 0.1, the **product $\sigma$-algebra** $\mathcal{S} \otimes \mathcal{T}$ is the $\sigma$-algebra on $S \times T$ generated by the measurable rectangles.

**Example 0.1** (Product $\sigma$-algebras)**.**

- Give the die rolls $D$ the $\sigma$-algebra $\mathcal{S}$ of all its subsets. Each single pair $\{(a, b)\} = \{a\} \times \{b\}$ in $D \times D$ is a measurable rectangle. Every subset of $D \times D$ is a union of finitely many single pairs, and $\mathcal{S} \otimes \mathcal{S}$ contains finite unions of its sets (Theorem 0.1), as well as $\emptyset$, so $\mathcal{S} \otimes \mathcal{S}$ is the collection of all subsets of $D \times D$.
- The rectangle $[0, 2] \times [0, 3]$ is a measurable rectangle for the Borel $\sigma$-algebras, so it is in $\mathcal{B} \otimes \mathcal{B}$.

**Definition 0.3** (Product measure)**.** Let $(S, \mathcal{S}, \mu)$ and $(T, \mathcal{T}, \nu)$ be measure spaces with $\sigma$-finite measures $\mu$ and $\nu$. The **product measure** $\mu \otimes \nu$ is the measure on the product $\sigma$-algebra $\mathcal{S} \otimes \mathcal{T}$ that gives each measurable rectangle the product of the measures of its sides:

$$(\mu \otimes \nu)(A \times B) = \mu(A)\, \nu(B) \quad \text{for all } A \in \mathcal{S} \text{ and } B \in \mathcal{T},$$

with $0 \cdot \infty = \infty \cdot 0 \stackrel{\text{def}}{=} 0$ and $x \cdot \infty = \infty \cdot x \stackrel{\text{def}}{=} \infty$ for $x > 0$. Exactly one measure has these values [2], Theorem 18.2.

**Example 0.2** (Product measures)**.**

- For the counting measure $\mu$ on the die rolls $D$, $(\mu \otimes \mu)(A \times B) = |A||B|$, the number of pairs in $A \times B$. For example, the set of pairs in which both rolls are even is $\{2, 4, 6\} \times \{2, 4, 6\}$, with

$$\begin{aligned}(\mu \otimes \mu)(\{2, 4, 6\} \times \{2, 4, 6\}) &= 3 \cdot 3 \\ &= 9\end{aligned}$$

  pairs.

- For Lebesgue measure $\lambda$ on $\mathbb{R}$,

$$\begin{aligned}(\lambda \otimes \lambda)([0,2] \times [0,3]) &= 2 \cdot 3 \\ &= 6,\end{aligned}$$

the area of the rectangle.

## Further reading

- Billingsley [2] develops measure theory and builds probability on it. Its early chapters define $\sigma$-algebras, measures, and their basic properties.
- Folland [21] is a graduate text on measure and integration; its first chapters treat $\sigma$-algebras and measures.
- Tao [22] is a one-quarter graduate introduction to measure and integration theory that assumes only an undergraduate analysis course and builds the theory through exercises. It treats $\sigma$-algebras, measures, and Lebesgue integration.
- Gut [23] is a graduate course in probability that opens with a chapter on introductory measure theory.

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